

UNDERGRADUATE COMMITTEE REPORTS: 2015

Undergraduate Committee Reports

Faculty Meeting, January 20, 2015

1. Undergraduate recruiting update

2015: 151 applications, 79 admitted, 1 accepted

2014: 231 applications, 88 admitted, 2 accepted (Feb 2014)

2013: 140 applications, 34 admitted, 3 accepted

Fall 2015 Admitted Student Statistics (79 total):

35 Female/ 44 Male

60 IL, 13 Non-IL, 6 International

28 White, 19 Asian, 17 Hispanic, 6 African-American, 4 International, 4 Multi-race, 1 unknown

Average ACT score: 29.3- High: 36 Low 25

2. Recruitment/Enrollment initiatives and schedule

a. Follow-up communication with students applying:

i. Incomplete applicants (application deadline extended to Feb 16th)

1. Send Email: by January 30th

ii. Admitted students → Intent to Enroll Deadline May 1st

1. Welcome email (AZ/LN): by February 6th
2. Phone calls (KM/AZ)- March 16th – April 3rd
3. Final email (AZ/LN): April 24th

b. Community college recruitment (Transfer student deadline March 31st)

i. **Moraine Valley CC**, College of DuPage, Harper CC, Wright CC, Oakton CC

1. Contact Guidance Counselors, Math/Science Chairs for possible visits during February/March

3. Update on teaching/lab equipment:

a. Process control system; installation, set-up, implementation

- i. Pressure & level module set up & working, software installed
- ii. PLC and PID controller setup in progress
- iii. Incorporate lab experiments in Process Control course → Unit Ops Fall 2015

b. Distillation column;

- i. Delivered mid-December; moved into unit ops lab January 13th.
- ii. Uncrating, setup and facility upgrade in progress.
- iii. Initial column run-through: Mid-March

4. ABET rubrics

- a. Faculty: Reminder to complete rubric for sample of students in your courses (30 students); deadline February 15 2015

5. COE Knowles Women's Summer Program

- a. Proposal to COE due January 30th
- b. Participating faculty to send AZ module information by Friday January 23rd

Draft Chemical Engineering Learning Module:

24-hour learning module/curriculum consisting of 2-hour modules, 1 per day for 12 days (maybe 2 hours in AM or 2 hours in PM).

Traditional ChE applications (Armfield demos)	4 hours	Zdunek
Nano-sensors	12 hours	Berry
Bio-engineering	4 hours	Akpa
Soft materials	4 hours	Sharma

Proposal Information:

- 1) Project name & objective
- 2) Curriculum breakdown (short syllabus)
- 3) Class location
- 4) Budget for materials, etc.
- 5) Number of grad students or TA's required to conduct the project
- 6) Any other additional information.

6. EPC meeting update:

- a. Nothing to report

Respectfully submitted,
Alan Zdunek
Director of Undergraduate Studies

Undergraduate Committee Report
Faculty Meeting, February 24, 2015

7. Undergraduate recruiting update

2015-Feb: 185 applications, 108 admitted, 4 accepted

2014-Feb: 231 applications, 107 admitted, 3 accepted

2013-Feb: 140 applications, 59 admitted, 6 accepted

Recruitment/Enrollment

- a. Follow-up communication with students:
 - i. Incomplete applicants: Reminder Emails sent/feedback
 - ii. Admitted students: Welcome Emails sent
 - 1. Phone calls - March 16th – April 3rd
 - 2. Final email: April 24th
- b. Women in Engineering Scholarship interviews held (2/14, 2/21) – 9 ChE's
- c. College of Engineering Celebratory Dinner for Admits (2/18) – 5 ChE's

8. Teaching/lab equipment:

- a. COE Annual Fund proposal
- b. Process control system; installation, set-up
- c. Distillation column set-up ongoing
- d. Classroom modules- heat exchangers

9. ABET rubrics

- a. Faculty: Reminder to complete rubric for sample of students in your courses (30 students); extended deadline (Spring break)

10. COE Knowles Women's Summer Program

- a. Chemical Engineering proposal sent out- 20 students have signed up.

11. Professional Development/ Career Initiatives

- a. Task force with UIC Career Center, 2 senior ChE students

12. EPC meeting update:

- b. Nothing to report

Respectfully submitted,
Alan Zdunek
Director of Undergraduate Studies

Undergraduate Committee Report: Faculty Meeting, March 17, 2015

13. Faculty vote on LAS changes to Math and Chemistry courses: Effective Fall 2016

- a. Summary of Proposed Changes (details on next page):
 - i. Math 180/181 → 5 to 4 credit hours (-2 credit hours)
Adding vectors, matrices to course curriculum
 - ii. CHEM 112 → CHEM 122/123; CHEM 114 → CHEM 124/125 (splitting lecture/lab)
(No change in course curriculum or credit hours for Gen Chem)
 - iii. CHEM 233 (Organic Lab) → 1 credit hour to 2 credit hours (+1 credit hour)
- b. CHE program impact:
 - i. “Electives Outside of Major Rubric” hours increased from 3 to 4
Issue- most courses available are 3 credit hours; to satisfy requirement, students would have to take 2 courses.
 - ii. “NonEng and Gen Ed degree requirements” hours decreased from 74 to 73.
- c. Recommendations:
 - Option 1:**
 - i. Keep “Electives Outside of Major Rubric” at 3 credit hours
 - ii. Change CHE 397 from 3 to 4 credit hours; Change Total Hours Required in College of Engineering to 49 hours
 - iii. “NonEng and Gen Ed degree requirements” hours decreased from 74 to 73.
 - Option 2**
 - i. Keep “Electives Outside of Major Rubric” at 3 credit hours
 - ii. Change ENGR100 from 0 to 1 credit hour; Change Total Hours Required in College of Engineering to 49 hours: : (Being discussed at COE level)
 - iii. “NonEng and Gen Ed degree requirements” hours decreased from 74 to 73.

14. ABET rubrics

- a. **REMINDER** to complete Fall 2014 rubric for sample of students in courses (30 students)

15. Senior Exit Interviews/Oral Exam

- a. DRAFT SCHEDULE for student interviews/oral exams in April:
 - i. April 6-10, April 13-17, April 20, 22, 24, April, 27, 29, 30
 - ii. **Please send me what dates/times work for you**
 - iii. Faculty will interview 5 students each; Alan(12 students)

16. Undergraduate recruiting update

2015-Mar: 220 applications, 150 admitted, 5 accepted
2014-Mar: 247 applications, 141 admitted, 4 accepted
2013-Mar: 211 applications, 111 admitted, 12 accepted
2012-Mar: 174 applications, 86 admitted, 13 accepted

Recruitment/Enrollment

- a. ITE students (Accepts): Thank you emails have been sent out
- b. 10 ChE Women in Engineering Scholarships: Congratulatory emails sent to students
 - i. 3 Unger awards, 1 Unger + Engineering Merit Awards, 1 Major + Engineering Merit Awards, 2 Dean's Scholarship Awards, 3 SWE Knowles Scholarships

17. Teaching/lab equipment:

- a. Process control system; ready for CHE 341
- b. Distillation column setup ongoing
- c. Classroom modules-
 - i. Tube/shell Heat exchanger shown to CHE201 class (no demo)
 - ii. Heat exchanger demo planned for CHE 312j

18. CHE Career Initiatives Task Force

- a. UIC Chemical Engineering LinkedIn account- existing since 2012 (Han Wang owner; contacted Han and he wants to help with updating LinkedIn acct and alumni/student outreach efforts)
- b. ChE alumni list with company contacts initiated by Rose Coppola-Conroy

Respectfully submitted,
Alan Zdunek, DUGS

LAS Justification for Changes

CHEM 112/ CHEM 114/CHEM 233

Separating the lecture and laboratory components of general chemistry courses into concurrent lecture and laboratory courses provides a better structure to support student progress and course accessibility. Currently, CHEM 112 (and 114) are comprised of a lecture and laboratory component; students complete a single registration and receive one grade. Because the courses award five hours of credit, poor performance in one component of a course (especially lecture) has a significant impact on the student's GPA; and the student who wishes to improve his/her grade must retake the entire course. With the revision, students who do well in, for example, the laboratory course, but not the lecture course, will not need to repeat both courses. This revision will also benefit the student's GPA because it will lessen the impact of a poor grade, while and permitting the inclusion of the student's better grade.

CHEM 233, Organic Chemistry, is a laboratory course that meets for 4 hours per week but awards only 1 hour of credit. There are no other chemistry courses with this many lab contact hours for only one hour of credit. The increase in credit hours to 2 will bring the course into consistency with the campus' credit hour/contact hour formula.

MATH 180/ MATH 181/ MATH 210

At most universities, the calculus I—calculus II sequence is, at most, 8 credit hours. Although the number of credit hours for Math 180 (and 181) will be reduced, the number of contact hours is unchanged and students will continue to have three hours of lecture and 2 hours of discussion each week. Discussion sections in Math 180 and Math 181 are problem sessions where students engage in active group learning projects under the direction of teaching assistants. As no additional material is covered in these sessions

it is reasonable that the number of credit hours is lower than the number of contact hours. At the request of stakeholder departments, the rudiments of linear algebra are being included in the first-year calculus sequence. A section on vectors will be added to Math 180 (and a section on matrices to Math 181). The addition of this material should help students in courses like Math 220 (Differential Equations) and many courses in the physical sciences and engineering. To make room for this material, some of the most technical material on techniques of integration is being removed.

Math 210 previously had 1 hour per week of laboratory-discussion devoted to computer methods in multivariable calculus. Changes in technology make it much easier to incorporate this material seamlessly in the lectures. At the same time, the department feels that students were suffering by not having enough time devoted to problem solving. For this reason, the lab-discussion is being changed to a problem session where students engage in active group learning activities under the direction of a teaching assistant.

Undergraduate Committee Report: Faculty Meeting, April 14, 2015

1. - ABET course rubrics from last fall- Please send to me
2. - ABET course surveys for end of this semester
 - a. Revised php files; one last check and will send out to faculty
3. - ABET senior exit survey and senior exit interview/oral exam (new procedure)
 - a. Senior exit interviews will be given as a written exam without the oral exam portion. Previous oral exam questions were revised. The exam will be assigned to all the students. Randomly pick twenty students to grade using the rubric.
4. - Status of ChE curriculum change in response to LAS course changes (Fall 2016)
 - a. Sent our general acceptance and recommendation to Dana Wright, Director of Academic Program Development about changing CHE 397 to 4 credit hours to make up the one credit hour lost from Math changes. Dana's response- These changes may require some internal discussion and/or approval within the college. I will take a look and mock up a proposal to assist the department. That way we'll have something formal to share with the college's EPC (if needed) and with SCEP (if needed).
5. - Update on new teaching lab equipment
 - a. Heat exchanger module demo in CHE 312, Process Control lab using PCT 50 equipment ongoing this week, distillation column; Walter/John time needed to complete electric/water/venting work.
6. - Student enrollment data and S-STEM Scholarship offers

Enrollment update (see graphs for details)

2015-Apr: 245 applications, 151 admitted, 21 accepted

2014-Apr: 268 applications, 151 admitted, 10 accepted

2013-Apr: 239 applications, 120 admitted, 20 accepted

2012-Apr: 195 applications, 94 admitted, 23 accepted

7. GPIIP Chemical Engineering students: 9 total

Agata Chmiel

Demetrios Galanos

Arun Joseph

Christopher Mecinski

Amir Saeed

David Schulz

Krupa Shah

Aisha Torres

Veronica Villanueva

8. AIChE Chicago Section UIC McCormack Award recipient: Vincent Errichiello

9. CHE Career Initiatives Task Force

- a. Phone conference with Han Wang: Ideas- Chemical Engineering Industry Night with alumni (next Fall). To be planned by Rose Coppola-Conroy. LinkedIn Account moved over to UIC ChE Department.

Respectfully submitted,
Alan Zdunek, DUGS

Undergraduate Committee Report: Faculty Meeting, August 28, 2015

1. Enrollment Update (8/18/15):
 - a. Total UG: 278
 - b. Male: 173 Female: 105
 - c. Minority: 63 (Hispanic, African-Am, Multi-race)
 - d. Frosh: 28 Soph: 40 Junior: 65 Senior: 145
 - e. Biochemical Concentration: 31
 - f. ChE + Minor: 16 (CS, Bio, Econ, EE, CME, Bus, Math, ME, Mus, Phys)
2. Feedback on GPIP CHE interns:, Arun Joseph (LEW), Christopher Mecinski (VB), Krupa Shah (VS), Aisha Torres (BC), Veronica Villanueva (YL), Agata Chmiel (MIE), Demetrios Galanos (Office of Sustainability), Amir Saeed (internship abroad), David Schulz (Office of Sustainability)
3. 1st Annual ChE Industry Day Event (October 23rd): ChE Department and Engineering Career Office
 - a. Concurrent sessions: A-1) Alumni panel with students A-2) Alumni/faculty lunch
 - b. Full session: Alumni/students/faculty.
4. UIC ChE LinkedIn Group: 78 members (ChE alumni): Invite current students, faculty, industrial board.
5. EPC Update: Submitting CHE 311 CRS (MATH 220 pre-requisite) at next week's meeting, LAS changes to be discussed.
6. Fall Recruitment: UIC Open House (Saturday, October 3rd, ChE student volunteers needed), Two Preview Days (September 12, October 11th), Firestarter Event December 5th (Early Action Admission)
7. ABET Rubrics: Please send to me by September 18th

Respectfully submitted,
Alan Zdunek, DUGS

Undergraduate Committee Report: Faculty Meeting, September 25, 2015

1. Enrollment Update (9/14/15):
 - a. Total UG: 289
 - b. Male: 181 Female: 108
 - c. Minority: 65 (Hispanic, African-Am, Multi-race)
 - d. Frosh: 28 Soph: 37 Junior: 70 Senior: 154
2. 1st Annual ChE Industry Day Event (October 23rd): ChE Department and Engineering Career Office
3. UIC ChE LinkedIn Group: 124 members, alums, faculty, students.
4. EPC Update: Preparation for Administration Council Meeting (focus on undergraduate education)
5. Fall Recruitment: UIC Open House (Saturday, October 3rd, 6 ChE student volunteers, need a couple more.

Coll	Major	First Year	Transfer	Guest	Total
ENGR	Bioengineering	36	6	49	91
	Chemical Engineering	21	5	38	64
	Civil Engineering	24	9	52	85
	Computer Engineering	36	9	41	86
	Computer Science	49	18	62	129
	Electrical Engineering	9	11	22	42
	Engineering Management	6	3	15	24
	Engineering Physics	12	0	14	26
	Industrial Engineering	3	1	5	9
	Mechanical Engineering	31	6	45	82
	Undecided – Focus on Engineering	2	1	2	5
	ENGR Totals	229	69	345	643

6. AIAA (American Institute of Aeronautics & Astronautics) UIC Student Chapter: Matt Kubik (ChE) Pres.; request to collaborate on rocket competition (liquid propulsion: kerosene/O₂)
7. UG Committee recommendation; Dept subscription to Chemical Engineering Education journal
8. Professional Development money for faculty (yearly) from COE

Respectfully submitted,
Alan Zdunek, DUGS

Undergraduate Committee Report: Faculty Meeting, October 27, 2015

1. 1st Annual ChE Industry Day Event (October 23rd)
 - 18 employers attended – of those employers, 13 alumni
 - Faculty/employer lunch feedback?
 - Student session: 82 registered students, 54 attended
 - Freshmen – 2
 - Sophomore – 3
 - Juniors – 15
 - Seniors – 34
2. Fall Recruitment:
 - a. UIC Open House (Saturday, October 3rd) 35 prospective CHE students
 - b. Preview session (October 12th) 2 prospective CHE studentsFollow-up emails sent
 - c. Next preview session, Wednesday, November 11th
3. Honors College Chemistry course change: UG committee to meet and make recommendations

Respectfully submitted,

Alan Zdunek, DUGS

Undergraduate Committee Report: Faculty Meeting, December 1, 2015

1. Honors College Chemistry change (CHEM 116/118): UG committee recommendations (handout)
2. CHE 499 Professional Development course:
 - a. UG committee recommends moving course to Fall semester starting Fall 2016
3. Minor in Chemical Engineering revision (handout):
 - a. Remove CHEM 342 from prerequisite course list
 - b. Add CHE 201 and CHE205 to required courses
4. Undergraduate Application Update: Spring/Fall 2016
 - a. Spring 2016:
 - Applications: 46
 - Admitted: 17
 - Intent to Enroll (ITE): 11
 - b. Fall 2016: (handout)

Respectfully submitted,

Alan Zdunek

Sultana Ferdous

Michael Caracotsios